COL1101 Lecture 19 2026/09/03

Topic: Binary Heaps

Announcements

Recap

Level Order Traversal: Visit nodes in increasing order of depth; within each level, left to right.

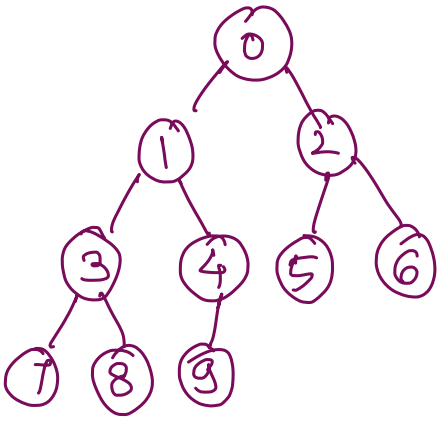
Complete Binary Tree (CBT): All levels full.

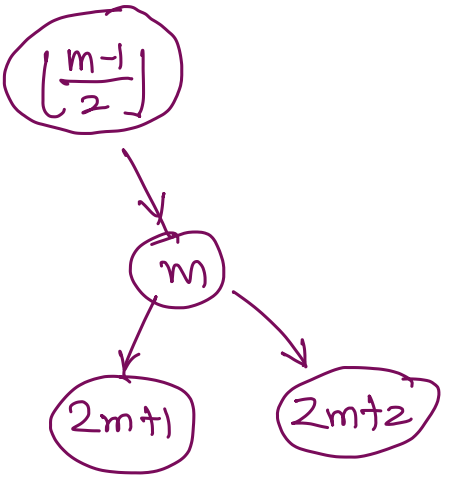
Almost Complete Binary Tree (ACBT): All levels except bottommost full. Nodes in bottommost level "as much to the left as possible".

Claim: Let n be the # of nodes in an almost complete binary tree and h be its height. Then $2^{h-1} \leq n \leq 2^h - 1$. Equivalently, $h = \lceil \log_2(n+1) \rceil$.

Construct all possible almost complete binary trees having level-order traversal

0 1 2 3 4 5 6 7 8 9

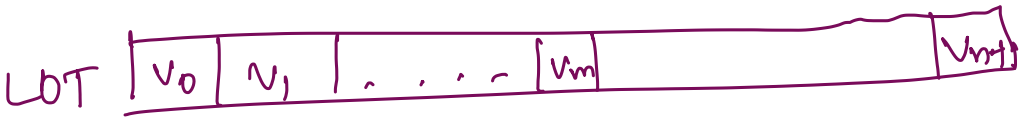




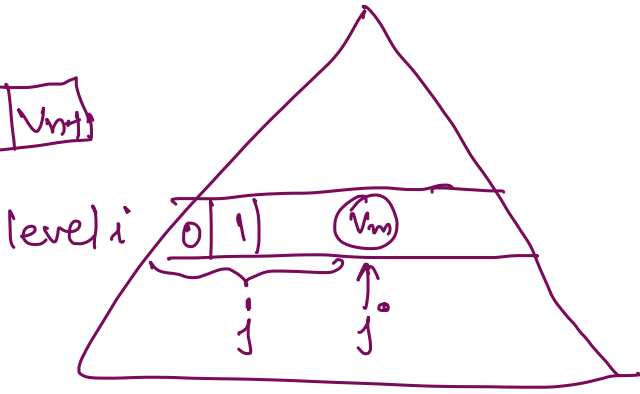
Claim: Let $V_0, V_1, \ldots, V_{n-1}$ be the level-order traversal of an ACBT. Then

- the parent of V_m is $V_{\lfloor \frac{m-1}{2} \rfloor}$ (provided $m > 0$).
- the left child of V_m is V_{2m+1} (provided $2m+1 < n$ o/w V_m doesn't have a left child).
- the right child of V_m is V_{2m+2} (provided $2m+2 < n$ o/w V_m doesn't have a right child).

Proof: For each level, number the nodes from left to right starting from 0.



What is the relation between m, i, j?



nodes on levels $0, \ldots, i-1$ =

$1 + 2 + \ldots + 2^{i-1} = 2^i - 1$

$\therefore m = \underbrace{2^i - 1}_{\text{upper levels}} + \underbrace{j}_{\text{same level, but left}} + 1 - \underbrace{1}_{\text{because we numbered LOT starting from 0.}} = 2^i - 1 + j$

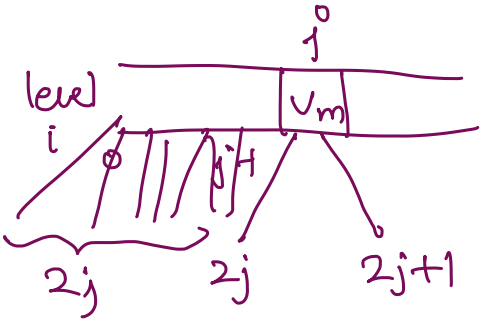
The children of v_m are on level $i+1$

Left child of v_m is v_{i+1}

$2^{i+1} - 1 + 2j$

$= v_{2m+1}$

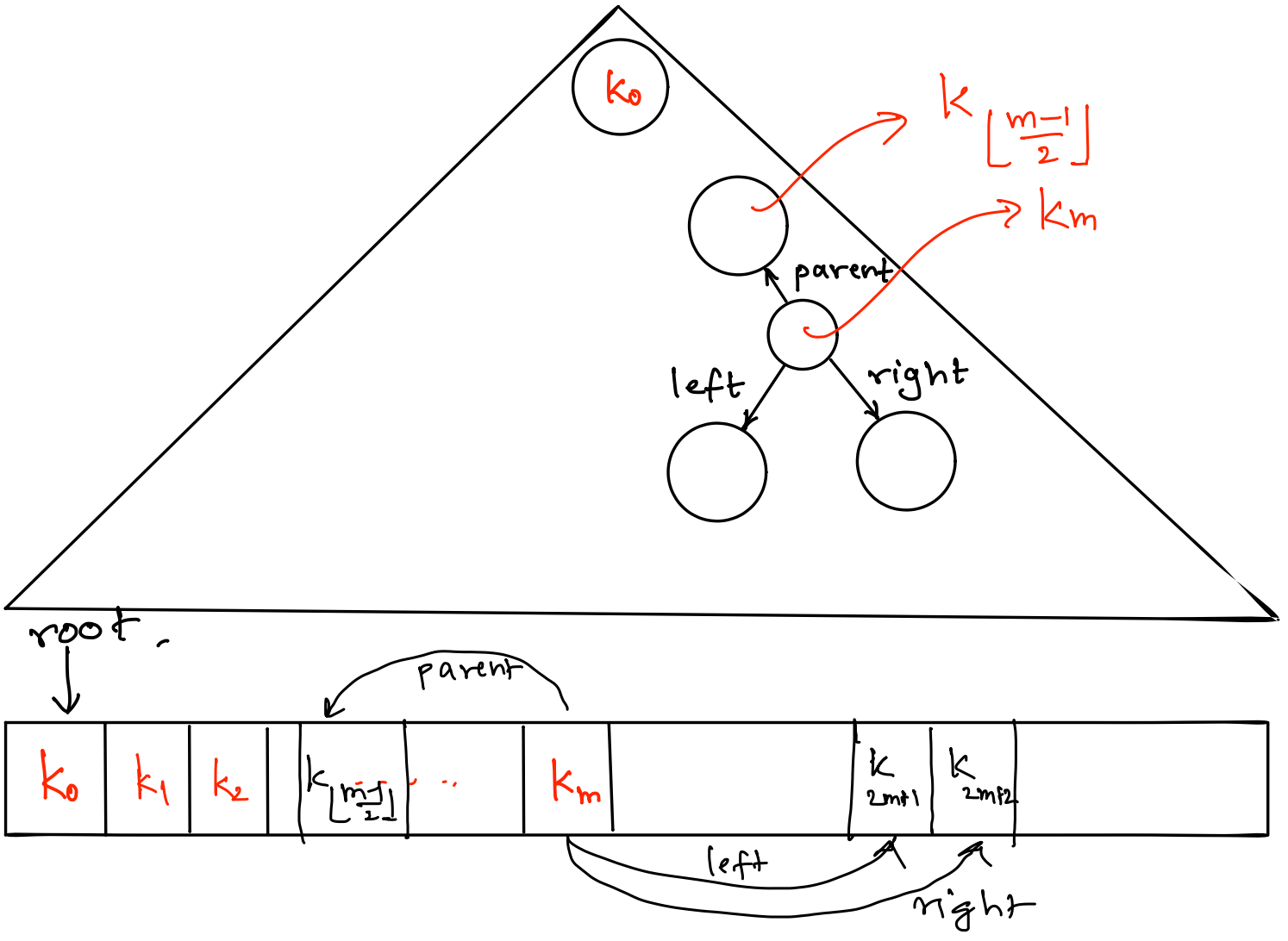
$\therefore$ Right child of v_m is v_{2m+2}.



Corollary: There exists a unique almost complete binary tree having any given sequence as its level order traversal.

Array / List representation of Almost-Complete Binary Trees

Represent an almost-complete binary tree by its level-order traversal



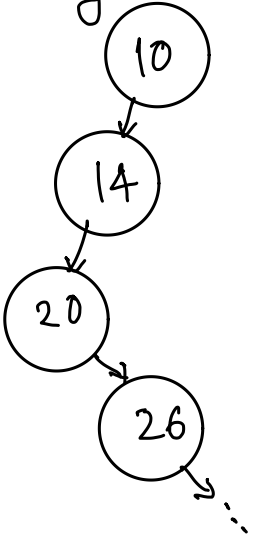
Set of keys in a priority queue:

{92, 32, 79, 14, 65, 38, 46, 10, 35, 26, 20, 89}

Sorted order:

[10, 14, 20, 26 32, 35, 38, 46, 65, 79, 89, 92]

Binary tree representation(?)

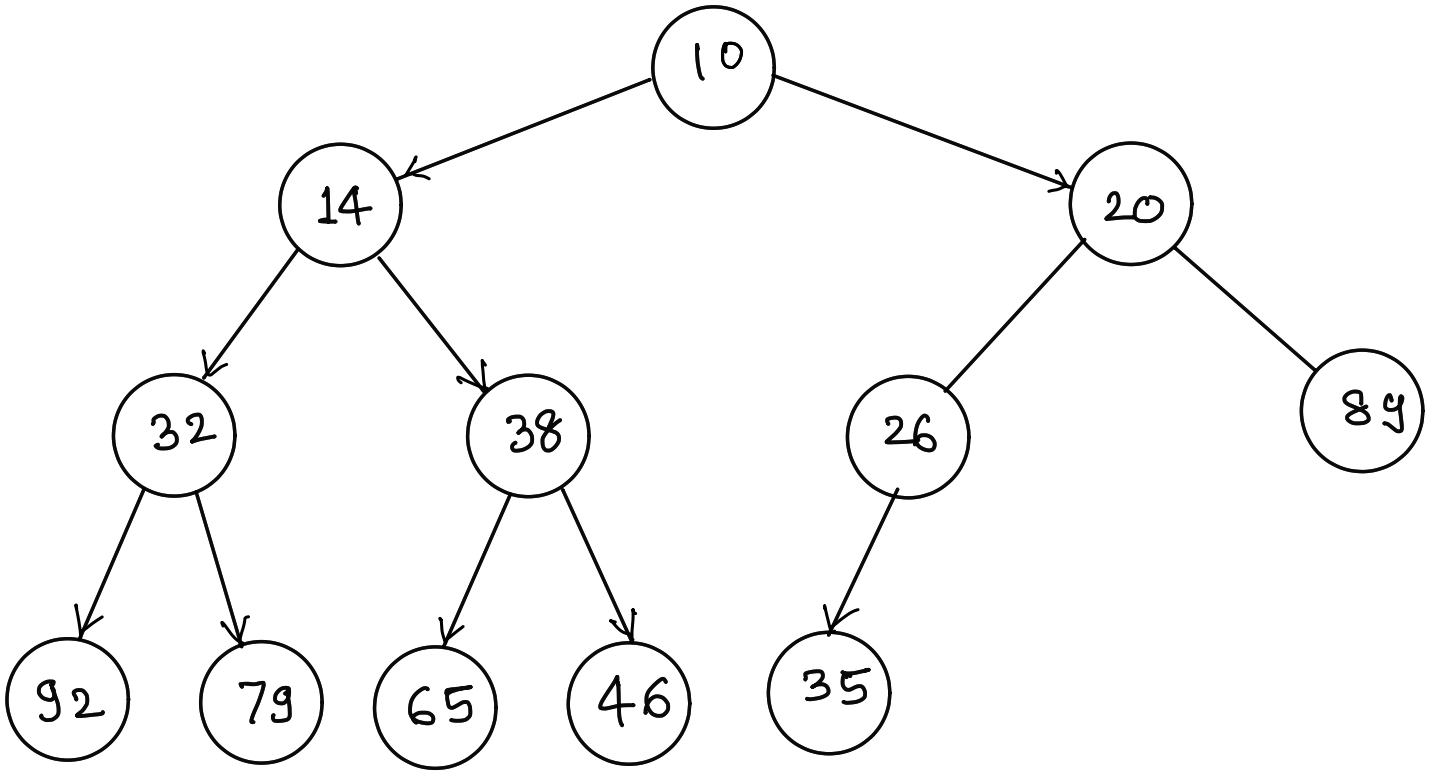


v.parent.key ≤ v.key for every v ≠ root.

ExtractMin: easy; shift root.

Enqueue: $\Omega(n)$ in the worst case;
height of the tree is n.

ACBT representation:



Binary Heaps

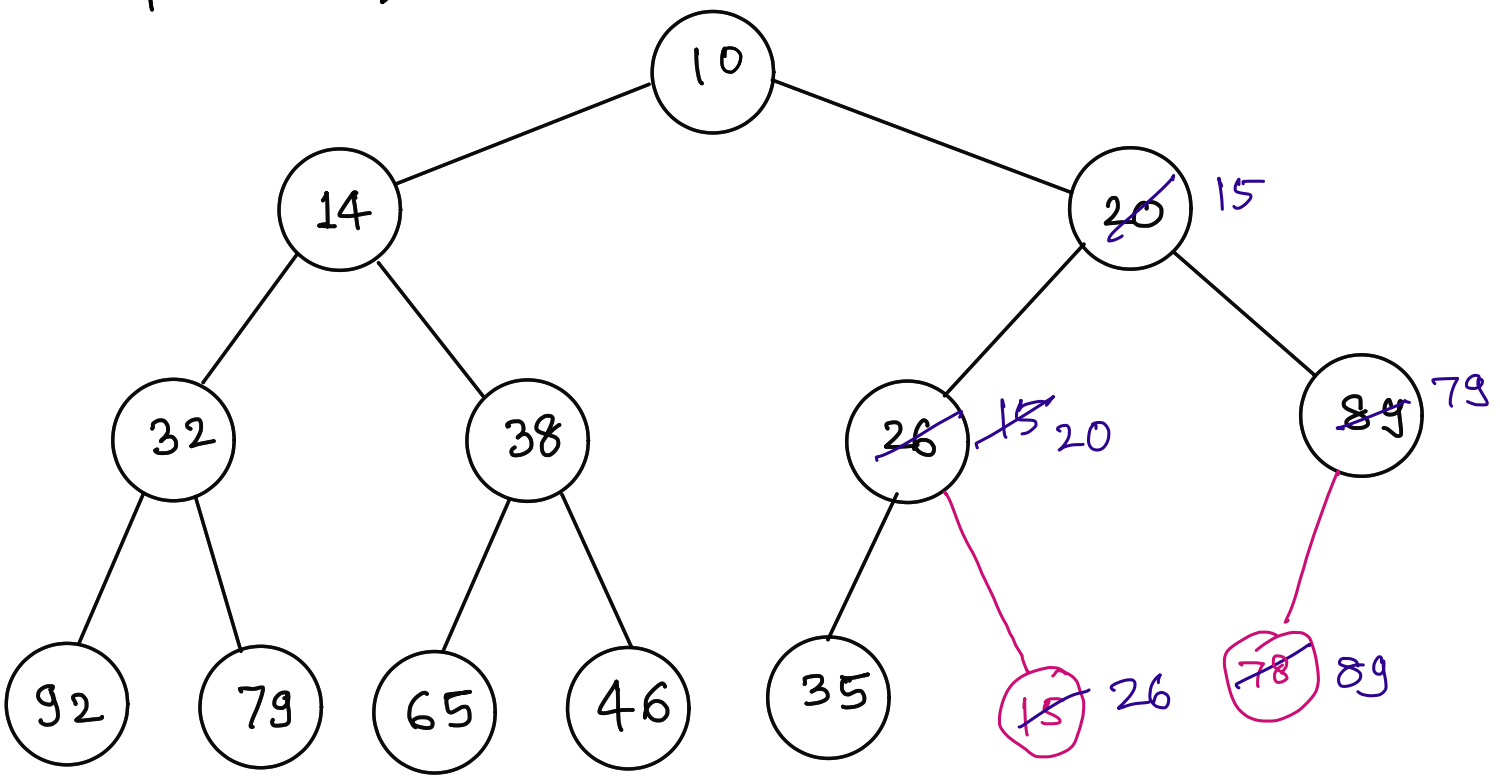
Definition: Let $(S, \leq)$ be an ordered set. A binary tree T with keys from S is called a binary min-heap if it satisfies the following two properties.

- [Structure] T is an ACBT
- [Order] For every node v of T which is not the root of T, we have: $v.parent.key \leq v.key$.

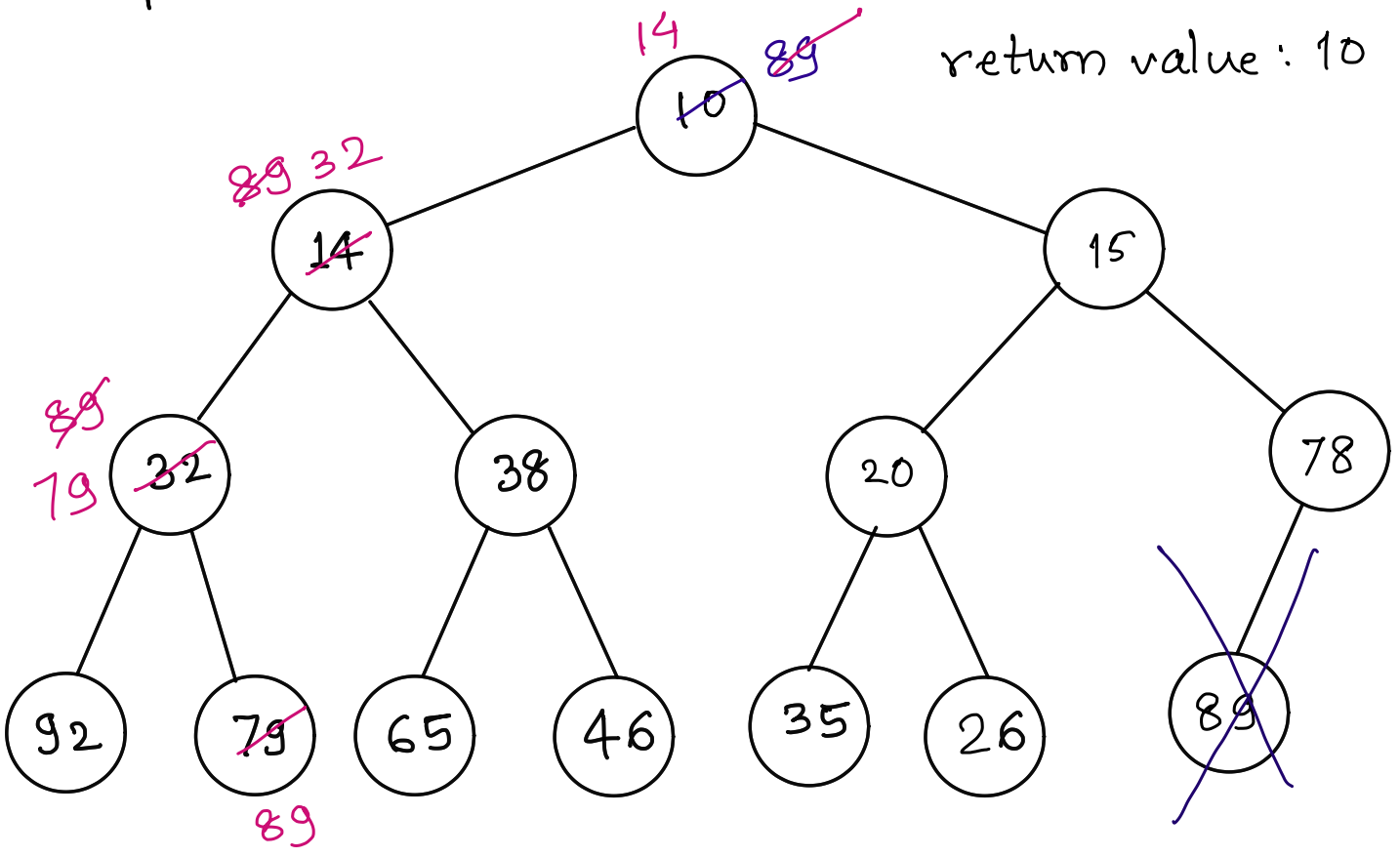
(Binary max-heap defined analogously.)

Proposal: Implement priority queue ADT using binary min-heaps.

Example: Enqueue (15), then Enqueue (79)



Example: ExtractMin()



Worst case running time of both Enqueue, ExtractMin is $O(h) = O(\log n)$.

Proof of correctness: next class.